Dawson Metzger-Fleetwood
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EDUCATION

University of Maryland — **College Park, MD**
B.S., Computer Science with Machine Learning Specialization — December 2022
B.S., Finance

SKILLS

Programming: Swift, Embedded Swift, SwiftUI, UIKit, RealityKit, TypeScript, React, HTML/CSS/JS, PHP, Python, C interop
Software: Xcode, VSCode, Node/Bun, Docker, Github Actions, TestFlight/Xcode Cloud, Jira, AWS (CDK, ECS/Fargate, Lambda, SQS, DynamoDB, S3, Bedrock, CloudFront, Cognito, SES), Fly.io, HuggingFace, PyTorch, MCP, Claude Code, Codex, Cursor

WORK EXPERIENCE

About Objects — **Reston, VA** | July 2023 - Present
AI Engineer

- Architected and built a Swift agent harness for iOS and visionOS with on device and cloud models, screen aware dynamic tool loading, context engineering through progressive disclosure, and voice control with interruption handling and under 100 ms time to first audio.
- Evaluated and rebuilt an AI document ingestion pipeline using AWS CDK infrastructure and autoscaling ECS/Fargate workers. Supports 100+ page documents with higher fidelity. Benchmarked 4× faster at one eighth the model API cost.
- Worked closely with leadership to rebuild the company's SDLC around AI assisted delivery, establishing impact metrics, an eval driven self improving review feedback loop, and a company wide skills repository that embeds engineering and project management workflows into agent behavior.
- Led development of a 3D ML object recognition pipeline, enabling rotation inference from 200 images. Supported an industrial oil drill manufacturer in analyzing drill bit component history and repairs.

Software Engineer

- Developed a multi-platform (iPad, iPhone, and Vision Pro) event management platform for government clients, featuring real-time resource tracking and 3D visualizations with RealityKit. Featured in Apple's enterprise showcase.
- Scaled the platform from inception to support 10,000+ concurrent users, built real time collaboration across distributed command teams, and reduced hierarchical rendering latency by 80%.
- Scaled a CI/CD pipeline with GitHub Actions and automated TestFlight builds as the team quadrupled in size. Mentored junior engineers and interns.
- Led the company website overhaul with custom WordPress plugins and a PHP backend.

Johns Hopkins University — **Baltimore, MD**
Guest Lecturer and Consultant — Spring 2024 - Present

- Designed accessible training materials and presented to interpreters on statistical concepts to enhance research interpretation, covering topics from fundamentals to the central limit theorem and multiple linear regression.
- Led development of applications designed to integrate internal systems and improve preparation efficiency, including a web app and MCP server for learning and rehearsing domain-specific vernacular ahead of technical talks, and a remote tooling server exposed over the internet via secure Cloudflare tunnels with OAuth 2.0.

Visual Language Associates, Inc — **Silver Spring, MD**
Technical Co-Founder — September 2019 - July 2023

- Co-founder of an agency that provides signed language interpreting and cued language transliteration services.
- Secured contracts with the MD and VA state governments, serving over two dozen clients in the DMV area with an average contract value of over 90,000. Grew the business from founding to several million dollars in revenue.
- Led the automation and integration of client payment, payroll, and scheduling software to streamline operations.
- Developed methodology to screen employees, clients, and contracts, as well as match clients with service providers.

ADDITIONAL INFO

Languages: German (proficient), American Sign Language (proficient), Cued English (fluent)

Open Source

AWS Amplify Swift SDK (github) — Summer 2026

- Diagnosed and fixed a visionOS build failure in AWS Amplify Swift's WebAuthn passkey support.

BookPlayer SharePlay Support (github) — Summer 2026

- Added SharePlay support to BookPlayer, a popular open source audiobook player (4.8/5 from 18K+ App Store ratings).

Personal Projects — **Silver Spring, MD**

Gemma 4 MoE Fine-Tuning (huggingface.co/dawsonamf) — Spring 2026

- Fine-tuned Google's Gemma 4 26B MoE model on an H100 using arbitrary rank ablation. Benchmarked two Pareto-optimal variants against Google's eval suite (MMLU, AIME 2026, GPQA Diamond, etc). Published on Huggingface.

Embedded Swift Agent (live demo, github) — Fall 2025

- Built a CLI coding agent from scratch in Embedded Swift with C interop resulting in a 200 KB binary (macOS, Linux, WASM) and 120ms cold start. Supports parallel tool execution, MCP client, subagents, web search, etc.

SiliconFly (github, writeup) — Summer 2026

- Built a sparse Metal kernel for a 139,255 neuron simulation, achieving 13.5× real time performance on an M4 Pro; validated results against an independent CPU reference.